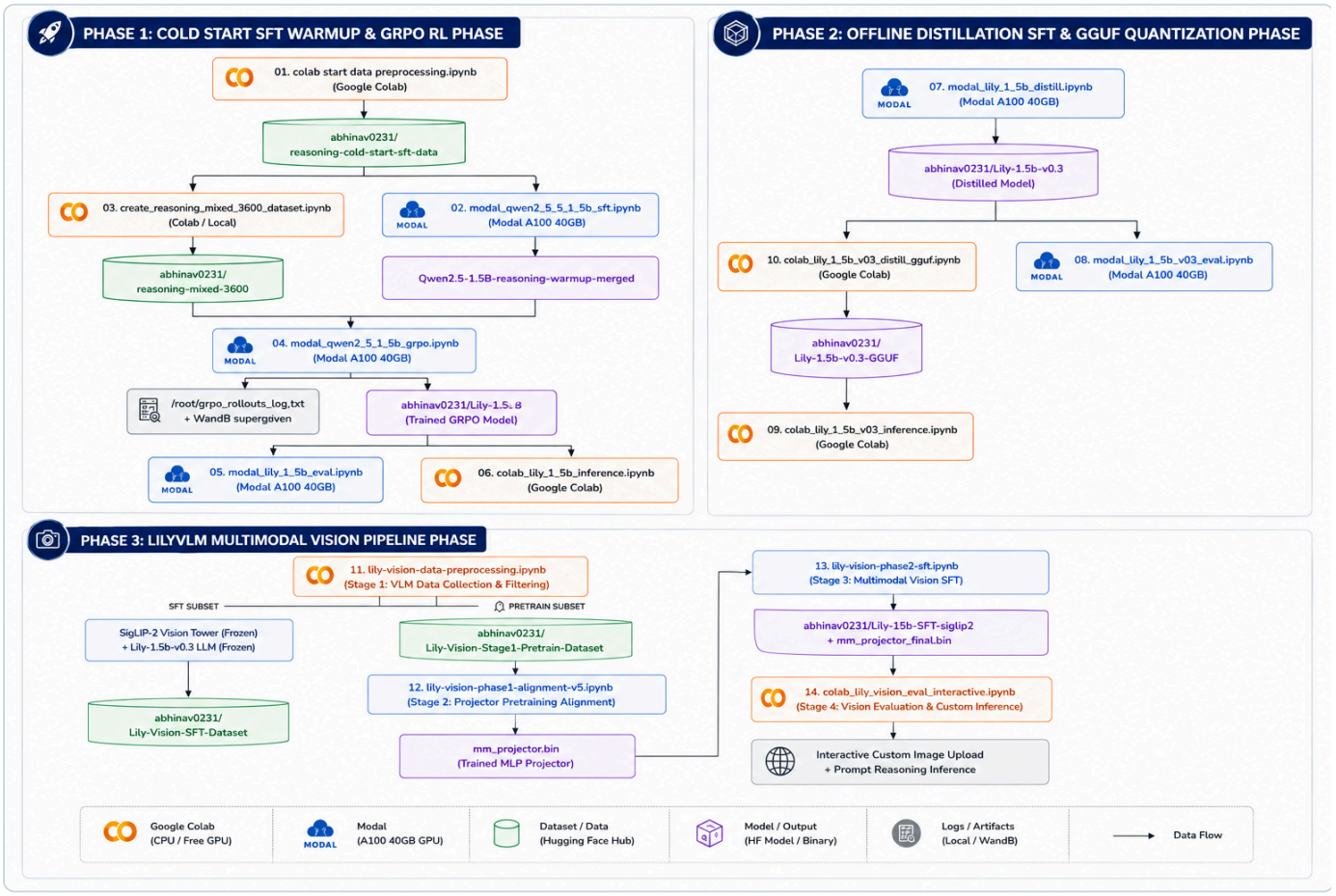
AI Training Handbook

Executive Summary & Engineering Architecture

This handbook serves as the definitive technical reference manual for our LLM and Vision-Language Model (VLM) training ecosystem. Synthesized directly from the production code across all 13 notebooks in our codebase, it bridges theoretical deep-learning concepts with real-world implementation technicalities.



1. Fine-Tuning Foundations: PEFT & LoRA Technical Deep-Dive

1.1 Mathematical Formulation of Low-Rank Adaptation (LoRA)

In standard full fine-tuning, an LLM parameter matrix $W_0 \in \mathbb{R}^{d \times k}$ is updated directly via gradient descent ($W = W_0 + \Delta W$). When d and k are large (e.g., hidden dimension $d = 1536$ and intermediate MLP dimension $k = 8960$ in Qwen2.5-1.5B), updating ΔW directly requires massive VRAM for storing optimizer states (AdamW requires 8 bytes per parameter for first and second moments).

Explicit Variable Definitions:

- $W_0 \in \mathbb{R}^{d \times k}$: Pre-trained frozen base weight matrix of the target linear layer.
- d: Input feature dimension (in-features / hidden size, e.g., $d = 1536$ in Qwen2.5-1.5B).
- k: Output feature dimension (out-features / projection size, e.g., $k = 8960$ in SwiGLU MLP layers).
- $\Delta W \in \mathbb{R}^{d \times k}$: High-dimensional weight update matrix resulting from fine-tuning gradients ($W = W_0 + \Delta W$).

- r: Inner low-rank hyperparameter ($r \ll \min(d, k)$).
- $A \in \mathbb{R}^{r \times k}$: Down-projection adapter matrix, initialized with Gaussian distribution $\mathcal{N}(0, \frac{1}{r})$.
- $B \in \mathbb{R}^{d \times r}$: Up-projection adapter matrix, initialized to zero ($B = 0$) ensuring $\Delta W = 0$ at step 0.
- α: Constant LoRA scaling hyperparameter. The ratio $\frac{\alpha}{r}$ scales the adapter's contribution relative to the base model.

LoRA parametrizes the weight update matrix ΔW by decomposing it into two low-rank matrices:

$$\Delta W = B \cdot A$$

During forward propagation, the linear layer output h for an input x is computed as:

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r}(B \cdot Ax)$$


Weight update in regular finetuning vs LoRA

1.2 QLoRA: NormalFloat4 (NF4) Base Weight Quantization

In QLoRA, the base model weights W_0 are quantized to 4-bit NormalFloat (NF4) precision, while the adapter matrices A and B remain in 16-bit precision (BF16 or FP16). NF4 is an information-theoretically optimal quantile quantization scheme for normally distributed weights:

$$q_i = \frac{1}{2}\left(Q_X\left(\frac{i}{2^k}\right) + Q_X\left(\frac{i+1}{2^k}\right)\right)$$

where $Q_X(\cdot)$ is the quantile function of the standard normal distribution $\mathcal{N}(0, 1)$.

1.3 Target Module Definitions & Code Implementation

Across our codebase (`02_modal_qwen_2_5_1_5b_sft.ipynb` , `04_modal_qwen_2_5_1_5b_grpo.ipynb` , `07_modal_lily_1_5b_distill.ipynb` , `13_lily-vision-phase2-sft.ipynb`), we apply LoRA across all 7 linear projection layers of the Transformer architecture:

```
from unsloth import FastLanguageModel

model = FastLanguageModel.get_peft_model(
    model,
    r = 32,                 # Rank: 32 (SFT/Distill/Vision), 64 (GRPO RL)
    lora_alpha = 64,         # Alpha: 32 (SFT), 64 (GRPO/Distill/Vision)
    target_modules = [
        "q_proj", "k_proj", "v_proj", "o_proj",    # Self-Attention Projections
        "gate_proj", "up_proj", "down_proj"         # SwiGLU MLP Projections
    ],
    lora_dropout = 0,        # 0 is optimized for Unsloth kernel fusion
    bias = "none",
    use_gradient_checkpointing = "unsloth",
    random_state = 42,
)
```

Detailed Technical Explanation of Target Modules:

- `q_proj` **(Query Projection)**: Transforms input hidden states into Query matrices (Q) for multi-head self-attention.
- `k_proj` **(Key Projection)**: Transforms input hidden states into Key matrices (K) for computing attention dot-product scores (QK^T).
- `v_proj` **(Value Projection)**: Transforms input hidden states into Value matrices (V) carrying contextual token information.
- `o_proj` **(Output Projection)**: Projects concatenated multi-head attention outputs back to the model hidden dimension ($d = 1536$).

- `gate_proj` **(Gate Projection in SwiGLU)**: Computes the element-wise gating signal in SwiGLU feed-forward networks ($\text{Swish}(xW_{\text{gate}}) \otimes xW_{\text{up}}$).
- `up_proj` **(Up Projection in SwiGLU)**: Expands input hidden dimension to high-dimensional intermediate space ($1536 \to 8960$).
- `down_proj` **(Down Projection in SwiGLU)**: Contracts intermediate dimension back to model hidden dimension ($8960 \to 1536$).

Trainable Parameter Efficiency Calculation

For Qwen2.5-1.5B ($d_{\text{model}} = 1536, N_{\text{layers}} = 28$): - Total Base Parameters: $1,580,643,840$ (~1.58B) - Trainable Adapter Parameters ($r = 32$): $36,929,536$ (~36.9M) - **Trainable Ratio**: $\frac{36.9\text{M}}{1580.6\text{M}} \approx 2.34\% \text{ to } 4.0\%$

1.4 Lossless 16-Bit Adapter Merging

Before running RL (GRPO) or exporting to GGUF, adapter weights must be mathematically merged into the base model to eliminate dual-path inference overhead:

$$W_{\text{merged}} = W_0 + \frac{\alpha}{r}(B \cdot A)$$

In code (`02` , `04` , `07` , `13`):

```
model_m, tok_m = FastLanguageModel.from_pretrained(
    model_name     = OUTPUT_DIR,
    max_seq_length = MAX_SEQ_LENGTH,
    dtype          = torch.bfloat16,  # Full precision BF16 merge
    load_in_4bit   = False,           # Prevents quantization artifacts during merge
)

model_m.save_pretrained_merged(MERGED_DIR, tok_m, save_method="merged_16bit")
model_m.push_to_hub_merged(HF_MERGED_REPO, tok_m, save_method="merged_16bit", token=HF_TOKEN)
```

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2. Hyperparameter Engineering & Optimization Strategy

2.1 The Global Effective Batch Size & Step Calculation

1. What is a Training Step?

A **single optimizer step** corresponds to one iteration where model weights are updated by the optimizer using calculated gradients.

2. Micro-Batches vs. Gradient Accumulation

To fit large models into GPU memory, training datasets are split into smaller **micro-batches** (B_{device}). When gradient accumulation is enabled (N_{accum}), the engine performs N_{accum} forward and backward passes, accumulating gradients in VRAM, before executing a single `optimizer.step()` update.

3. Mathematical Formula for Effective Batch Size:

$$B_{\text{eff}} = B_{\text{device}} \times N_{\text{accum}} \times N_{\text{GPU}}$$

where: - B_{device}: Per-device micro-batch size processed in a single forward pass per GPU. - N_{accum}: Number of micro-batches accumulated before updating model weights. - N_{GPU}: Number of parallel GPUs participating in training.

4. Mathematical Formula for Total Optimizer Steps:

$$\text{Total Optimizer Steps} = \frac{|\mathcal{D}| \times N_{\text{epochs}}}{B_{\text{eff}}}$$

where $|\mathcal{D}|$ is the number of samples in the dataset and N_{epochs} is the total number of training passes over the dataset.

Concrete Worked Step Calculation Examples from Codebase:

- **SFT Warmup (Notebook 02)**: 4,000 samples $\times$ 2 Epochs = 8,000 total samples. $B_{\text{eff}} = 8 \times 2 \times 1 = 16$. $\text{Total Steps} = \frac{8000}{16} = 500$ steps.
- **Distillation (Notebook 07)**: 91,457 raw samples packed into 33,297 contiguous sequences $\times$ 2 Epochs = 66,594 sequence passes. $B_{\text{eff}} = 24 \times 1 \times 1 = 24$. $\text{Total Steps} = \frac{66594}{24} = 2,776$ steps.
- **Vision Phase 2 SFT (Notebook 13)**: 64,000 samples $\times$ 1 Epoch. $B_{\text{eff}} = 24 \times 2 \times 1 = 48$. $\text{Total Steps} = \frac{64000}{48} = 1,333$ steps.

2.2 Verified Production Hyperparameter Comparison Matrix

Hyperparameter	SFT Warmup (02)	GRPO RL (04)	Distillation (07)	Vision Phase 1 (12)	Vision Phase 2 (13)
Base Model	Qwen2.5-1.5B-Instruct	Qwen2.5-Warmup-Merged	Lily-1.5b-v0.1	Lily-1.5b-v0.3	Lily-1.5b-v0.3
Dataset Size ($\mathcal{D}$)	4,000 samples	3,600 prompts	91,457 (train)	121,237 samples	64,000 samples
Epochs / Steps	2 Epochs	600 Max Steps	2 Epochs (2,776 steps)	1 Epoch	1 Epoch
Per-Device Batch (B_{device})	8	4	24	16	24
Grad Accum (N_{accum})	2	4	1	3	2
Effective Batch (B_{eff})	**16**	**16**	**24**	**48**	**48**
Peak Learning Rate ($\eta_{\max}$)	1×10^{-5}	5×10^{-6}	2×10^{-5}	2×10^{-4}	2×10^{-5}
LR Scheduler	Cosine	Cosine	Cosine	Cosine w/ warmup	Cosine w/ warmup
Warmup Ratio	0.05	0.05	0.03	0.05	0.03
Optimizer	`adamw_torch_fused`	`adamw_torch_fused`	`adamw_torch_fused`	`AdamW` ($\beta_1 = .9, \beta_2 = .95$)	`AdamW`
Weight Decay	0.01	0.0	0.0	0.05	0.01
Max Sequence Length	3,072	3,072	4,096	256	3,072
Sequence Packing	False	False	**True**	False	False
Hardware	Modal A100 40GB	Modal A100 40GB	Modal A100 40GB	Modal A100 40GB	Modal A100 40GB

2.3 Learning Rate Schedules & Mathematical Formulation

We utilize Cosine Annealing with Warmup across all training runs.

$$\eta_t = \begin{cases} \eta_{\max} \cdot \frac{t}{T_{\text{warmup}}}, & t \leq T_{\text{warmup}} \\ \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{\pi(t - T_{\text{warmup}})}{T_{\text{total}} - T_{\text{warmup}}}\right)\right), & t > T_{\text{warmup}} \end{cases}$$


Cosine Learning Rate Schedule with Warmup

2.4 Optimizer Selection: `adamw_torch_fused` vs `adamw_8bit`

Across our primary high-performance training runs, we utilize `adamw_torch_fused`: 1. `adamw_torch_fused`: Fuses point-wise AdamW parameter and momentum update steps into a single CUDA kernel invocation. This eliminates intermediate GPU memory read/write cycles between PyTorch operators, maximizing GPU compute utility on A100. 2. `adamw_8bit` **(bitsandbytes)**: Quantizes 32-bit Adam first-moment (m_t) and second-moment (v_t) states to 8-bit block-wise representations, reducing optimizer memory overhead by 75%.

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3. Reinforcement Learning via GRPO (Group Relative Policy Optimization)

3.1 Mathematical Theory & Advantage Calculation

Group Relative Policy Optimization (GRPO) simplifies reinforcement learning by sampling a group of G candidate outputs $\{y_1, y_2, \ldots, y_G\}$ for each input prompt x from the current policy $\pi_{\theta_{\text{old}}}$.

For each completion y_i, GRPO evaluates a composite scalar reward score R_i and computes baseline-normalized advantages across the sampled group:

$$A_i = \frac{R_i - \mu(R_1, \ldots, R_G)}{\sigma(R_1, \ldots, R_G) + \epsilon}$$

where μ is the group reward mean, σ is the standard deviation, and $\epsilon = 10^{-4}$ ensures numerical stability.

3.2 DAPO Loss Objective with Asymmetric Clipping

The model parameters θ are updated by minimizing the Dynamic Advantage Policy Optimization (DAPO) clipped objective:

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|y_i|}\sum_{t=1}^{|y_i|}\left[\min\left(r_{i,t}(\theta)A_i,\ \text{clip}(r_{i,t}(\theta), 1-\epsilon_{\text{low}}, 1+\epsilon_{\text{high}})A_i\right)\right] + \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})$$

where: - **Probability Ratio**: $r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})}$ - **Asymmetric Clipping bounds**: $\epsilon_{\text{low}} = 0.2$ and $\epsilon_{\text{high}} = 0.28$. Asymmetric clipping allows the policy to take larger gradient steps when discovering positive high-advantage completions while restricting negative updates. - **Schulman Per-Token KL Divergence Penalty**:

$$D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) = \frac{\pi_{\text{ref}}(y_{i,t} \mid x, y_{i,<t})}{\pi_\theta(y_{i,t} \mid x, y_{i,<t})} - \log\frac{\pi_{\text{ref}}(y_{i,t} \mid x, y_{i,<t})}{\pi_\theta(y_{i,t} \mid x, y_{i,<t})} - 1$$

3.3 Detailed Variable Breakdown of `GRPOConfig` Parameters

```
from trl import GRPOConfig, GRPOTrainer

training_args = GRPOConfig(
    output_dir                  = "/root/grpo_output",
    num_generations             = 8,
    max_prompt_length           = 1024,
    max_completion_length       = 1536,
    temperature                 = 0.9,
    top_p                       = 0.95,
    learning_rate               = 5e-6,
    lr_scheduler_type           = "cosine",
    warmup_ratio                = 0.05,
    max_steps                   = 600,
    per_device_train_batch_size = 4,
    gradient_accumulation_steps = 4,
    loss_type                   = "dapo",
    epsilon                     = 0.2,
    epsilon_high                = 0.28,
    beta                        = 0.001,
    mask_truncated_completions  = True,
    bf16                        = True,
    optim                       = "adamw_torch_fused",
)
```

Complete Narrative Parameter Descriptions:

- `output_dir`: The target directory path on disk where intermediate LoRA checkpoints, training logs, and merged models are saved.
- `num_generations` **($G = 8$)**: The group size G. Controls how many independent output completions are sampled per input prompt to compute relative group advantage A_i.
- `max_prompt_length` **(1024)**: Truncates user input prompts exceeding 1024 tokens to conserve VRAM during generation.
- `max_completion_length` **(1536)**: Caps generated rollout responses at 1536 tokens, providing sufficient budget for deep Chain-of-Thought (CoT) reasoning.
- `temperature` **(0.9)**: Sampling temperature used during rollout generation. A value of 0.9 encourages diverse exploration of reasoning strategies.
- `top_p` **(0.95)**: Nucleus sampling threshold restricting sampling to the top 95% cumulative probability token mass.
- `learning_rate` **(5×10^{-6})**: Peak learning rate for policy gradient updates. Set conservatively low to maintain RL policy stability.
- `lr_scheduler_type` **("cosine")**: Decay schedule decreasing the learning rate smoothly following a cosine curve over 600 steps.
- `warmup_ratio` **(0.05)**: Linear warmup phase covering the first 5% of training steps (30 steps).
- `max_steps` **(600)**: Total number of RL optimizer steps executed during training.
- `per_device_train_batch_size` **(4)**: Micro-batch size of prompts loaded into VRAM per GPU generation call.
- `gradient_accumulation_steps` **(4)**: Accumulates gradients across 4 micro-batches ($4 \times 4 = 16$ effective prompt batch size).
- `loss_type` **("dapo")**: Enables Dynamic Advantage Policy Optimization with asymmetric ratio clipping.
- `epsilon` **(0.2)**: Lower clipping threshold ϵ_{low} limiting negative ratio updates.
- `epsilon_high` **(0.28)**: Upper clipping threshold ϵ_{high} allowing positive advantage rewards to scale further.
- `beta` **(0.001)**: Scaling coefficient for the KL divergence penalty term $D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})$.

- `mask_truncated_completions` **(True)**: Prevents incomplete outputs that reach `max_completion_length` without an EOS token from corrupting policy gradients.
- `bf16` **(True)**: Enables 16-bit BFloat16 execution on NVIDIA Ampere GPUs.
- `optim` **("adamw_torch_fused")**: Fused AdamW optimizer used for fast step updates.

3.4 In-Depth Technical Explanations of the 4 Custom Reward Functions

Our GRPO pipeline utilizes 4 domain-aware reward functions to guide model behavior across structure, factual accuracy, instruction compliance, and sequence length.

Reward 1: `format_reward` (Structural CoT & Tag Enforcement)

- **Technical Purpose**: Enforces strict XML structural tags (`\lt think>...\lt /think>` and `\lt answer>...\lt /answer>`) and penalizes model responses that skip Chain-of-Thought reasoning.
- **Scoring Mechanics**:
- Checks if `\lt think>` and `\lt /think>` tags are both present. If present, awards $+0.20$.
- Extracts text inside `\lt think>...\lt /think>`. If reasoning length is ≥ 50 words, awards an additional $+0.15$ for substantive reasoning. If missing, penalizes -0.15.
- Checks if `\lt answer>` and `\lt /answer>` tags are present. Awards $+0.15$ for tags and $+0.10$ if answer content is non-empty. Penalizes -0.15 if missing.

Reward 2: `correctness_reward` (Domain-Routed Correctness)

- **Technical Purpose**: Evaluates factual correctness based on the ground-truth target and domain `answer_type`.
- **Scoring Mechanics**:
- **Numeric Domain (`numeric`)**: Extracts numerical values using regular expressions `\d+(\.\d+)?` and compares prediction against ground truth within a tolerance of 1×10^{-3} (also evaluating fraction equivalences like $1/2 = 0.5$). Awards $+1.0$ if correct, -0.5 if wrong.
- **Exact / MCQ Domain (`exact`)**: Extracts candidate uppercase letters (`A` , `B` , `C` , `D`) for multiple-choice questions. Awards $+1.0$ for exact match, -0.5 for incorrect option selection.
- **Boolean Domain (`bool`)**: Normalizes responses to `"yes"` or `"no"`. Awards $+1.0$ for matching truth, -0.5 otherwise.
- **Code Domain (`code`)**: Analyzes python code structure for function declarations (`def`), return statements (`return`), and 4-space indentation. Awards proportional scores up to $+0.60$.

Reward 3: `instruction_reward` (Alpaca Word Count Adherence)

- **Technical Purpose**: Ensures model compliance with explicit formatting instructions (such as target word counts) in instruction prompts.
- **Scoring Mechanics**:
- Filters samples originating from the `alpaca` source.
- Parses target word count constraints from prompts using regex (e.g., "*in 50 words*").
- Calculates length difference $|N_{\text{actual}} - N_{\text{target}}|$. Awards $+1.0$ if within 12% tolerance, -0.5 if constraint is violated.

Reward 4: `length_reward` (Degenerate Output Guard)

- **Technical Purpose**: Serves as a guardrail preventing policy degeneration into single-word cop-outs or infinite repetitive loops.
- **Scoring Mechanics**:
- Computes total word count W.
- If $W < 15$ words (under-reasoning): penalizes -0.5.

- If $W > 2500$ words (runaway generation): penalizes -0.25.
- If $15 \leq W \leq 2500$ words: awards $+0.15$ bonus.

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4. Technical Interpretation of ALL GRPO Training Metrics & Plots

Monitoring Weights & Biases (W&B) diagnostic plots is crucial during GRPO reinforcement learning to verify training health and detect policy anomalies.


GRPO RL Training Diagnostics

4.1 Detailed Breakdown of the 6 Diagnostic Training Panels

Panel 1: DAPO Policy Loss & Optimization Stability

- **What it Measures**: Plots the DAPO policy gradient loss $\mathcal{L}_{\text{GRPO}}$ across 600 training steps.
- **How to Interpret**: Healthy training displays a sharp drop in loss during the first 100 steps as the model aligns with format requirements, followed by bounded oscillation between $[-0.1, +0.2]$. Persistent positive loss values (> 2.0) signal gradient instability or an aggressive learning rate.

Panel 2: KL Divergence $D_{\text{KL}}(\pi_\theta || \pi_{\text{ref}})$ & Policy Drift

- **What it Measures**: Tracks Kullback-Leibler divergence between current policy π_θ and frozen base reference model π_{ref}.
- **How to Interpret**: Normal policy updates remain bounded within $0.001 \leq D_{\text{KL}} \leq 0.05$. The red dashed line illustrates **Policy Collapse / Reward Hacking**, where KL divergence spikes exponentially as the model degenerates to exploit reward loopholes.

Panel 3: Total Mean Reward Trajectory

- **What it Measures**: Displays total composite reward score $R_{\text{total}} = R_{\text{format}} + R_{\text{correctness}} + R_{\text{instruction}} + R_{\text{length}}$.
- **How to Interpret**: Demonstrates monotonic growth from initial negative scores (-0.2) up toward target performance ceiling ($+1.5+$), confirming effective policy learning.

Panel 4: Component Rewards Breakdown

- **What it Measures**: Deconstructs composite reward into individual reward curves (R_{format}, $R_{\text{correctness}}$, R_{length}).
- **How to Interpret**: `format_reward` (blue curve) saturates near maximum (+0.60) within 40 steps, while `correctness_reward` (green curve) grows steadily over 600 steps as complex reasoning improves.

Panel 5: Reasoning Chain Length Dynamics

- **What it Measures**: Tracks average generated token length per response across steps.
- **How to Interpret**: Shows dynamic expansion of token count from ~210 tokens up to ~950 tokens as the model spontaneously develops step-by-step Chain-of-Thought (CoT) reasoning inside `\lt think>` tags.

Panel 6: Gradient Norm $||g||_2$ & Stability

- **What it Measures**: Measures the L_2 norm of model parameter gradients before optimizer step execution.
- **How to Interpret**: High initial gradient norms (> 4.0) stabilize below 1.0. Gradient clipping (`max_grad_norm=1.0`) prevents destabilizing weight updates.

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5. Offline Distillation & Loss Function Deep-Dive

5.1 Offline Distillation Mechanics

In our pipeline, offline distillation transfers reasoning capabilities from a 105-Billion parameter teacher model (`Sarvam-105b-Distill-100k`) to our 1.5-Billion parameter student model (`Lily-1.5B`).

Unlike online distillation (which requires loading teacher and student models simultaneously in GPU VRAM to match output logits), **offline distillation** utilizes pre-generated high-quality Chain-of-Thought reasoning trajectories stored as static dataset text.

```
[ Sarvam 105B Teacher ] ---> Pre-computes 100k CoT Trajectories ---> HF Hub Dataset
                                                                          |
[ Lily 1.5B Student ] \lt --- Supervised Fine-Tuning (CE Loss) \lt ------------+
```

5.2 Token-Level Cross-Entropy Loss Formulation: Comprehensive Theory & Mechanics

1. What Does Cross-Entropy Loss Measure?

Cross-Entropy (CE) Loss measures the **information-theoretic discrepancy** (or negative log-likelihood) between the true target token distribution $P_{\text{target}}(y_t)$ (the teacher model's completion, represented as a 1-hot target vector over the vocabulary V) and the student model's predicted probability distribution $P_\theta(y_t \mid y_{<t}, x)$.

Mathematically, Cross-Entropy between true distribution P and predicted distribution Q is:

$$H(P, Q) = -\sum_{v \in V} P(v) \log Q(v) = H(P) + D_{\text{KL}}(P \| Q)$$

Since the ground-truth target is a deterministic one-hot vector ($H(P) = 0$), minimizing Cross-Entropy Loss is **mathematically identical to minimizing the KL divergence** $D_{\text{KL}}(P_{\text{teacher}} \| P_{\text{student}})$ between the teacher's distribution and the student's predictions!

2. How is Token-Level Cross-Entropy Loss Calculated Step-by-Step?

1. **Logit Computation**: The student model passes input context x and preceding tokens $y_{<t}$ through its Transformer layers to produce unnormalized logit vectors $z_t \in \mathbb{R}^{|V|}$ (where vocabulary size $|V| = 151,936$).
2. **Softmax Normalization**: Logits are converted into a probability distribution via Softmax:

$$P_\theta(y_t = k \mid y_{<t}, x) = \frac{\exp(z_{t,k})}{\sum_{j=1}^{|V|} \exp(z_{t,j})}$$

3. **Target Surprisal Calculation**: The loss for target token y_t is its negative log probability: $-\log P_\theta(y_t \mid y_{<t}, x)$.
4. **Label Masking (m_t)**: Prompt tokens are assigned `labels = -100` ($m_t = 0$), ensuring zero gradient computation over prompt tokens. Assistant completion tokens are assigned target token IDs ($m_t = 1$).
5. **Masked Sequence Loss Formulation**:

$$\mathcal{L}_{\text{CE}}(\theta) = -\frac{1}{\sum_{t=1}^{T} m_t} \sum_{t=1}^{T} m_t \cdot \log P_\theta(y_t \mid y_{<t}, x)$$

5.3 Sequence Packing Mechanics: Memory Layout & Efficiency Gains

1. The Problem with Naive Batch Padding

In standard PyTorch DataLoaders, sequences in a batch are padded with `\lt pad>` tokens to match the length of the longest sequence in that batch. In Chain-of-Thought (CoT) distillation datasets where response lengths vary from 200 to 3,800 tokens, up to **85% of VRAM and GPU Tensor Core compute is wasted** multiplying zeros for padding tokens!

2. How Sequence Packing Works (`packing=True` in Notebook 07)

Sequence Packing concatenates multiple individual text samples into a single contiguous token array of fixed length `MAX_SEQ_LENGTH = 4096`. Samples are separated by `\lt |im_end|>` special tokens.

```
Unpacked Batch (Wasteful Padding):
[Sample 1 (500 tokens) | \lt pad> \lt pad> ... (3596 pad tokens)] -> 87% VRAM Wasted!
[Sample 2 (1200 tokens)| \lt pad> \lt pad> ... (2896 pad tokens)] -> 70% VRAM Wasted!

Packed Sequence (100% Compute Efficiency):
[Sample 1 (500t) \lt |im_end|> Sample 2 (1200t) \lt |im_end|> Sample 3 (2394t)] -> 0% Wasted!
```

3. Attention Masking in Packed Sequences

To prevent attention leakage across concatenated samples inside a packed sequence, Unsloth utilizes **Block-Diagonal Attention Masking** (FlashAttention-2 cumulative sequence indexing `cu_seqlens`), ensuring token t in Sample 2 can only attend to preceding tokens within Sample 2.

4. Empirical Efficiency Outcomes:

From `07_modal_lily_1_5b_distill.ipynb` execution logs: - Raw Dataset: **91,457 individual text samples**. - Packed Dataset: **33,297 packed sequences of length 4,000**. - Throughput Boost: **>2.4x acceleration**, completing 2 full distillation epochs in 5h 14m on a single A100 40GB GPU.

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6. Model Evaluation, Benchmarking & Schema Verification

6.1 `lm-eval` Harness Execution & Configuration

In `05_modal_lily_1_5b_eval.ipynb` (v0.1) and `08_modal_lily_1_5b_v03_eval.ipynb` (v0.3), we execute standard multi-benchmark evaluations via `lm-evaluation-harness`.

Command Line Invocations (Modal A100 & L4)

```
# Notebook 08 (Lily 1.5b v0.3 Evaluation)
lm_eval \
  --model hf \
  --model_args pretrained=abhinav0231/Lily-1.5b-
v0.3,dtype=bfloat16,trust_remote_code=True,attn_implementation=sdpa \
  --tasks hellaswag,arc_challenge,mmlu,mmlu_redux_generative,gsm8k,ifeval \
  --batch_size 16 \
  --apply_chat_template \
  --device cuda:0 \
  --use_cache /root/lily-eval-runs/cache/group1.db \
  --output_path /root/lily-eval-runs/results/group1
```

Benchmark Overview

1. **GSM8K**: Grade school math word problems (measures multi-step numerical reasoning).
2. **ARC-Challenge**: AI2 Reasoning Challenge (grade-school science questions requiring logic).
3. **HellaSwag**: Sentence completion benchmark evaluating commonsense NLI.
4. **MMLU / MMLU-Redux**: 57 academic subjects (elementary math, physics, humanities, law).
5. **IFEval**: Instruction-Following Evaluation (evaluates formatting constraints like word counts, JSON format, bullet counts).

6.2 Schema Compliance Verification Framework

In `09_colab_lily_1_5b_v03_inference.ipynb`, we built a programmatic schema verification engine to audit whether quantized models maintain strict `\lt think>` and `\lt answer>` tag structures.

```
def extract_stats(text):
    return {
        "think_open":     text.count("\lt think>"),
        "think_close":    text.count("\lt /think>"),
        "answer_open":    text.count("\lt answer>"),
        "answer_close":   text.count("\lt /answer>"),
        "thinking_leak":  text.count("\lt thinking>"), # Audit tag leakage
        "im_end":         text.count("\lt |im_end|>")
    }

def classify_case(raw_stats):
    if raw_stats["think_open"] == 1 and raw_stats["think_close"] == 1 \
       and raw_stats["answer_open"] == 1 and raw_stats["answer_close"] == 1 \
       and raw_stats["thinking_leak"] == 0:
        return "PASS_exact"
    elif raw_stats["thinking_leak"] > 0:
        return "FAIL_thinking_leak"
    elif raw_stats["think_open"] == 1 and raw_stats["answer_open"] == 0:
        return "PARTIAL_only_think"
    else:
        return "FAIL_no_schema"
```

---

7. Model Export, Quantization & GGUF Conversion

7.1 GGUF Container Format & Quantization Theory

GGUF (GGML Unified Format) is a binary file format designed for fast, single-file model loading on edge devices (CPU/GPU via `llama.cpp`). Quantization reduces 16-bit float weight representations down to lower bit-widths (4-bit, 5-bit, 8-bit) using block-wise scaling factor quantization:

$$w \approx s \cdot q + m$$

where q is the quantized k-bit integer, s is the block scale factor, and m is the zero-point offset.

7.2 The 2 Export Pipelines in Our Codebase

Pipeline A: Unsloth Direct Export (`07_modal_lily_1_5b_distill.ipynb`)

```
# Exports GGUF files directly from Unsloth engine
model_m.save_pretrained_gguf("gguf_model", tok_m, quantization_method="q4_k_m")
```

Pipeline B: Manual `llama.cpp` CLI Pipeline (`10_colab_lily_1_5b_v03_distill_gguf.ipynb`)

1. **System Prompt Patching**: Replaces base Qwen system templates with our standardized CoT system prompt across `config.json` and tokenizer files.
2. **F16 GGUF Conversion**: `bash python llama.cpp/convert_hf_to_gguf.py \ /content/Lily-1.5b-v0.3 \ --outtype f16 \ --outfile /content/Lily-1.5b-v0.3-F16.gguf`
3. **K-Quantization Execution**: `bash llama.cpp/build/bin/llama-quantize /content/Lily-1.5b-v0.3-F16.gguf /content/Lily-1.5b-v0.3-Q4_K_M.gguf Q4_K_M llama.cpp/build/bin/llama-quantize /content/Lily-1.5b-v0.3-F16.gguf /content/Lily-1.5b-v0.3-Q5_K_M.gguf Q5_K_M llama.cpp/build/bin/llama-quantize /content/Lily-1.5b-v0.3-F16.gguf /content/Lily-1.5b-v0.3-Q8_0.gguf Q8_0`

Quantization Variant Comparison Matrix

Variant	Quantization Method	File Size (1.5B Model)	VRAM Required	Recommended Usage
`F16`	Unquantized FP16	3.09 GB	~4.2 GB	Baseline reference evaluation
`Q8_0`	8-bit Standard Quant	1.62 GB	~2.5 GB	Server-side high-precision inference
`Q5_K_M`	5-bit K-Quant (Medium)	1.12 GB	~1.8 GB	Optimal quality/memory balance
`Q4_K_M`	4-bit K-Quant (Medium)	0.98 GB	~1.5 GB	Mobile / Edge CPU deployment

---

8. Multimodal Vision Extension (Lily Vision)

8.1 VLM Architecture & Dimensionality Pipeline

Our Vision-Language Model **Lily Vision** (`11` , `12` , `13`) connects a pre-trained Vision Tower to the Lily 1.5B LLM via a 2-layer projection bottleneck:

```
# Projector Architecture Implementation (12_lily-vision-phase1-alignment-v5.ipynb)
class LilyVisionProjector(nn.Module):
    def __init__(self, vision_dim=1152, llm_dim=1536):
        super().__init__()
        self.downsampler = nn.AdaptiveAvgPool2d((18, 18)) # 18x18 = 324 tokens
        self.mlp = nn.Sequential(
            nn.Linear(vision_dim, 2048),
            nn.SiLU(),
            nn.Linear(2048, llm_dim)
        )

    def forward(self, x): # x: [B, 729, 1152]
        B, N, C = x.shape
        H = W = int(N ** 0.5) # 27x27 grid
        x_2d = x.transpose(1, 2).view(B, C, H, W)
        x_pooled = self.downsampler(x_2d).flatten(2).transpose(1, 2) # [B, 324, 1152]
        return self.mlp(x_pooled) # [B, 324, 1536]
```

8.2 Phase 1 Alignment vs Phase 2 SFT — Layer Freezing Matrix

Architecture Submodule	Phase 1: Projector Alignment (12)	Phase 2: Multimodal SFT (13)
SigLIP-2 Vision Encoder	❄️ **Frozen** (`requires_grad = False`)	❄️ **Frozen** (`requires_grad = False` , set to `.eval()`)
2-Layer MLP Projector	🔥 **Trained** (Random init, LR = 2×10^{-4})	🔥 **Trained** (From Phase 1 weights, LR = 2×10^{-5})
LLM Base Weights	❄️ **Frozen** (`requires_grad = False`)	❄️ **Frozen** (Loaded in 4-bit QLoRA)
LLM LoRA Adapters	N/A (No LoRA used)	🔥 **Trained** ($r = 32, \alpha = 64$ on 7 layers)
Visual Token Count	324 tokens (Pooled via 18x18 adaptive pool)	**729 tokens** (Full 27x27 patch resolution)
Sequence Length	256 tokens	3,072 tokens

8.3 `LengthGroupedBatchSampler` Performance Optimization

In `13_lily-vision-phase2-sft.ipynb` , multimodal sequences vary drastically in token length. Standard batching causes massive padding overhead. We implemented a custom PyTorch batch sampler that groups sequences by length:

```
class LengthGroupedBatchSampler(Sampler):
    def __init__(self, lengths, batch_size, mega_batch_mult=50, seed=42):
        self.lengths = lengths
        self.batch_size = batch_size
        self.mega_batch_size = batch_size * mega_batch_mult # 24 * 50 = 1200 samples

    def __iter__(self):
        # Partition into mega-batches, sort by sequence length, form mini-batches
        pass
```

Optimization Outcome: Reduced total Phase 2 SFT training duration from **~3 hours down to ~20 minutes** on an A100 40GB GPU.

---

9. Cloud Checkpointing & Fault-Tolerant Auto-Resumption

Across all cloud training notebooks (`02` , `04` , `07` , `12` , `13`), running on serverless infrastructure (Modal) presents a risk of preemption or timeout. We built custom Hugging Face Hub callbacks that automatically snapshot checkpoints to private repos and auto-resume on restart.

```
from huggingface_hub import HfApi, snapshot_download

# Auto-Resume Logic (07_modal_lily_1_5b_distill.ipynb)
if RESUME_FROM_CHECKPOINT and CHECKPOINT_REPO:
    api = HfApi()
    files = list(api.list_repo_files(CHECKPOINT_REPO, token=HF_TOKEN))
    ckpt_nums = [int(f.split("/")[0].split("-")[-1]) for f in files if "checkpoint-" in f]
    if ckpt_nums:
        latest = max(ckpt_nums)
        local_dir = os.path.join(OUTPUT_DIR, f"checkpoint-{latest}")
        print(f"Downloading latest checkpoint-{latest} from HF Hub...")
        snapshot_download(
            repo_id        = CHECKPOINT_REPO,
            allow_patterns = [f"checkpoint-{latest}/*"],
            local_dir      = OUTPUT_DIR,
            token          = HF_TOKEN,
        )
        _resume_ckpt = local_dir
```

Conclusion & Summary Checklist

By combining **PEFT (LoRA/QLoRA)**, **Hyperparameter Scaling Equations**, **GRPO Reinforcement Learning**, **Offline SFT Distillation**, **GGUF K-Quantization**, and **2-Phase Multimodal VLM Alignment**, our codebase establishes a complete, production-grade LLM development framework.